

Given: Due:	Further Maths (Core Pure) HW 07	Name:
Chapter 6 (6.1 – 6.4): Matrix arithmetic & 2x2 inverses		
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.		
91% (A*)	76% (A)	66% (B)
56% (C)	50% (D)	40% (E)
<i>Useful resources: Useful resources: Your notes from recent lessons & your text book</i> <u>Although most matrix arithmetic can be done on your calculator, show all steps to reach the answer & use calculator for checking results only.</u>		
Questions - do these on lined paper		
1) Given that k is a real number and that $\mathbf{A} = \begin{pmatrix} 1+k & k \\ k & 1-k \end{pmatrix}$ find the exact values of k for which $\mathbf{A}$ is a singular matrix. Give your answers in their simplest form. (3)		
2) Given that $\mathbf{M} = (\mathbf{A} + \mathbf{B})(2\mathbf{A} - \mathbf{B})$, $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}$ (a) calculate the matrix $\mathbf{M}$, (6) (b) find the matrix $\mathbf{C}$ such that $\mathbf{MC} = \mathbf{A}$. (4)		
3) (i) Given that $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \\ 4 & 5 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 2 & -1 & 4 \\ 1 & 3 & 1 \end{pmatrix},$ (a) find $\mathbf{AB}$. (b) Explain why $\mathbf{AB} \neq \mathbf{BA}$. (4)		
(ii) Given that $\mathbf{C} = \begin{pmatrix} 2k & -2 \\ 3 & k \end{pmatrix},$ where k is a real number find $\mathbf{C}^{-1}$, giving your answer in terms of k . (3)		
4) $\mathbf{M} = \begin{pmatrix} x & x-2 \\ 3x-6 & 4x-11 \end{pmatrix}$ Given that the matrix $\mathbf{M}$ is singular, find the possible values of x . (4)		
5)		



(a) Given that

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & 3 \\ 4 & 5 & 5 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 0 & -1 \end{pmatrix}$$

find $\mathbf{AB}$.

(2)

(b) Given that

$$\mathbf{C} = \begin{pmatrix} 3 & 2 \\ 8 & 6 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 5 & 2k \\ 4 & k \end{pmatrix}, \quad \text{where } k \text{ is a constant}$$

and

$$\mathbf{E} = \mathbf{C} + \mathbf{D}$$

find the value of k for which $\mathbf{E}$ has no inverse.

(4)

6)

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$$

(a) Show that $\mathbf{A}$ is non-singular.

(2)

(b) Find $\mathbf{B}$ such that $\mathbf{BA}^2 = \mathbf{A}$.

(4)

7)

$$\mathbf{M} = \begin{pmatrix} 2a & 3 \\ 6 & a \end{pmatrix}, \quad \text{where } a \text{ is a real constant.}$$

(a) Given that $a = 2$, find $\mathbf{M}^{-1}$.

(3)

(b) Find the values of a for which $\mathbf{M}$ is singular.

(2)

END OF HOMEWORK

Total = 41 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.